CSE 1321L: Programming and Problem Solving I Lab

Assignment 5 – 100 points

1D & 2D Arrays

What students will learn:
1) Declaring arrays
2) Initializing arrays
3) Performing common operations on arrays

Overview: Arrays are an incredibly powerful thing in computing. Almost every audio file, video file, and image you've ever seen on a computer is stored in an array. An array is simply a data structure that holds a lot of things of the same type (i.e., it is homogeneous). For example, they could hold 50 integers, 100 Booleans, or a million floats. For most languages, when you see brackets [ ], you know you're working with an array.

Assignment 5A: *Hospital Directory.* You've been asked to develop a directory look-up program for Community General Hospital. You will create multiple 1D arrays, predefined with the following information.

ID	First Name	Last Name	Age	Occupation	Has Medical License?
0	Mark	Sloan	67	Chief of Internal Medicine	TRUE
1	Amanda	Bentley	33	Pathologist	TRUE
2	Norman	Briggs	47	Administrator	FALSE
3	Steve	Sloan	35	Hospital Security	FALSE
4	Jesse	Travis	24	Surgeon Intern	TRUE

The ID column serves as a common index; the rest of the columns should be stored in individual arrays of the appropriate data type (especially important for the last column!).

You will then use a loop to create a menu that repeatedly gives the user the following choices:

- List all employee records (in order)

- Display an employee record by ID
  - The user should enter a valid ID. If the employee exists, print out their information. Otherwise, tell the user that it is an invalid ID

- Display an employee record by last name
  - The user should enter a valid last name. If employee(s) exist, print out their information. Otherwise, tell the user that there are no records.

- Quit

Sample Output:

```
[Community General Hospital Directory]
1) List all employees
2) Search employee records by ID
3) Search employee records by Last Name
4) Quit
Choice: 1

#0: Mark Sloan, Age 67, Chief of Internal Medicine, Medical License:
TRUE
#1: Amanda Bentley, Age 33, Pathologist, Medical License: TRUE
#2: Norman Briggs, Age 47, Administrator, Medical License: FALSE
#3: Steven Sloan, Age 35, Hospital Security, Medical License: FALSE
#4: Jesse Travis, Age 24, Surgeon Intern, Medical License: TRUE

1) List all employees
2) Search employee records by ID
3) Search employee records by Last Name
4) Quit
Choice: 2
ID: 10
Invalid ID!
ID: 1
#1: Amanda Bentley, Age 33, Pathologist, Medical License: TRUE

1) List all employees
2) Search employee records by ID
3) Search employee records by Last Name
4) Quit
Choice: 3
Last Name: Stewart
No Records Found!

1) List all employees
2) Search employee records by ID
3) Search employee records by Last Name
4) Quit
Choice: 3
Last Name: Sloan
#0: Mark Sloan, Age 67, Chief of Internal Medicine, Medical License:
TRUE
#3: Steven Sloan, Age 35, Hospital Security, Medical License: FALSE

1) List all employees
2) Search employee records by ID
3) Search employee records by Last Name
4) Quit
Choice: 5
No option 5!
Choice: 4
[Closing Directory...]
```

<u>Assignment 5B:</u> *Art Program.* We're going to build on our PBM image creator from Assignment 4 to create a fully functional image editor, using 2D arrays to store our changes. We'll also make it a little more interesting by switching to the Portable Gray Map standard. 8 bit PGM files have color values ranging from 0 (black) to 255 (white), with the values in-between being shades of gray. The header is very similar to PBM, and looks like this:

P2
WIDTH HEIGHT
255
IMAGE_DATA

The P2 tells image readers that it is a PGM file, and the 255 indicates the largest possible color value is 255. Otherwise, it's the same as Assignment 4.

You will first prompt the user to enter a width, height, and initial color value to fill the image. You should then create a 2D array with those dimensions and initialize each cell with that color value.

You will then use a loop to create a menu that repeatedly prompts the user to make one of the following choices:

- Fill in a pixel with a new color
  - Prompt the user to enter a row and column value and a new color value. If the array has that row and column index AND the color value is between 0 – 255, change the color at the index.
    Otherwise, inform the user that it's an invalid command.

- Fill in a line with a new color
  - Prompt the user to enter a starting row and column value, a new color value, a length value, and a direction (left, right, up, or down, case sensitive).
    If all those values are valid, fill in each index in the line with those values in the 2D array (you can open up a program like Excel if you need help visualizing this)
    If the row, column, color, and/or direction are not correct, inform the user that it's an invalid command.
    If the length is correct but would go outside the array (e.g. a length of 20 but the array only has 10 columns), fill in values up to the edge of the array and then stop. Do not cause an ArrayOutOfBounds error.

- Print the PGM file
  - This should print out the contents of a valid PGM image. You should be able to copy and paste the output into a file and load it in the same way as you did with the PBM file in Assignment 4 – the only difference is that the extension should be .p**g**m.

- Quit

<u>Sample Output:</u>

```
[KSU Image Manipulation Program]
Enter an image width: 5
Enter an image height: 7
Enter the fill color: 24
```

```
What will you do?
1) Fill in a pixel
2) Fill in a line
3) Print the image
4) Quit
Choice? 3

PGM Image Contents
P2
5 7
255
24 24 24 24 24
24 24 24 24 24
24 24 24 24 24
24 24 24 24 24
24 24 24 24 24
24 24 24 24 24
24 24 24 24 24

What will you do?
1) Fill in a pixel
2) Fill in a line
3) Print the image
4) Quit
Choice? 1

Row: 40
Column: 2
New Color: 23
Invalid information!
Row: 2
Column: 1
New Color: 123

What will you do?
1) Fill in a pixel
2) Fill in a line
3) Print the image
4) Quit
Choice? 2

Row: 4
Column: 3
New Color: 245
Length: 6
Direction: up
```

```
What will you do?
1) Fill in a pixel
2) Fill in a line
3) Print the image
4) Quit
Choice? 3

PGM Image Contents
P2
5 7
255
24 24 24 245 24
24 24 24 245 24
24 123 24 245 24
24 24 24 245 24
24 24 24 245 24
24 24 24 245 24
24 24 24 24 24

What will you do?
1) Fill in a pixel
2) Fill in a line
3) Print the image
4) Quit
Choice? 4

[Exiting KSU Image Manipulation Program]
```

Submission:

1. You will submit 2 separate files
2. File names and class names must be correct.
3. Upload all files (simultaneously) to the assignment submission folder in Gradescope.